

## Exercise 4.2

### Important Formulas

- (i)  $(x + y)^2 = x^2 + 2xy + y^2$
- (ii)  $(x - y)^2 = x^2 - 2xy + y^2$
- (iii)  $x^2 - y^2 = (x + y)(x - y)$
- (iv)  $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$
- (v)  $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$
- (vi)  $(x - y)^3 = x^3 - y^3 - 3xy(x - y)$
- (vii)  $(x + y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$
- (viii)  $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- (ix)  $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$

#### 1. Factorize each of the following expressions:

(i)  $4x^4 + 81y^4$

$$\begin{aligned} &4x^4 + 81y^4 \\ &= (2x^2)^2 + (9y^2)^2 + 2(2x^2)(9y^2) - 36x^2y^2 \\ &= (2x^2 + 9y^2)^2 - (6xy)^2 \\ &= (2x^2 + 9y^2 + 6xy)(2x^2 + 9y^2 - 6xy) \\ &= (2x^2 + 6xy + 9y^2)(2x^2 - 6xy + 9y^2) \end{aligned}$$

(ii)  $a^4 + 64b^4$

$$\begin{aligned} &a^4 + 64b^4 \\ &= (a^2)^2 + (8b^2)^2 + 2(a^2)(8b^2) - 16a^2b^2 \\ &= (a^2 + 8b^2)^2 - (4ab)^2 \\ &= (a^2 + 8b^2 + 4ab)(a^2 + 8b^2 - 4ab) \\ &= (a^2 + 4ab + 8b^2)(a^2 - 4ab + 8b^2) \end{aligned}$$

(iii)  $x^4 + 4x^2 + 16$

$$\begin{aligned} &x^4 + 4x^2 + 16 \\ &= x^4 + 16 + 4x^2 \\ &= (x^2)^2 + (4)^2 + 2(x^2)(4) - 8x^2 + 4x^2 \\ &= (x^2 + 4)^2 - 4x^2 \\ &= (x^2 + 4)^2 - (2x)^2 \\ &= (x^2 + 4 + 2x)(x^2 + 4 - 2x) \\ &= (x^2 + 2x + 4)(x^2 - 2x + 4) \end{aligned}$$

(iv)  $x^4 - 14x^2 + 1$

$$\begin{aligned} &x^4 - 14x^2 + 1 \\ &= x^4 + 1 - 14x^2 \\ &= (x^2)^2 + (1)^2 + 2(x^2)(1) - 2x^2 - 14x^2 \\ &= (x^2 + 1)^2 - 16x^2 \\ &= (x^2 + 1)^2 - (4x)^2 \\ &= (x^2 + 1 + 4x)(x^2 + 1 - 4x) \\ &= (x^2 + 4x + 1)(x^2 - 4x + 1) \end{aligned}$$

(v)  $x^4 - 30x^2y^2 + 9y^4$

$$\begin{aligned} &x^4 - 30x^2y^2 + 9y^4 \\ &= x^4 + 9y^4 - 30x^2y^2 \\ &= (x^2)^2 + (3y^2)^2 + 2(x^2)(3y^2) - 6x^2y^2 - 30x^2y^2 \end{aligned}$$

$$\begin{aligned} &= (x^2 + 3y^2)^2 - 36x^2y^2 \\ &= (x^2 + 3y^2)^2 - (6xy)^2 \\ &= (x^2 + 3y^2 + 6xy)(x^2 + 3y^2 - 6xy) \\ &= (x^2 + 6xy + 3y^2)(x^2 - 6xy + 3y^2) \end{aligned}$$

(vi)  $x^4 - 7x^2y^2 + y^4$

$$\begin{aligned} &x^4 - 7x^2y^2 + y^4 \\ &= x^4 + y^4 - 7x^2y^2 \\ &= (x^2)^2 + (y^2)^2 + 2(x^2)(y^2) - 2x^2y^2 - 7x^2y^2 \\ &= (x^2 + y^2)^2 - 9x^2y^2 \\ &= (x^2 + y^2)^2 - (3xy)^2 \\ &= (x^2 + y^2 + 3xy)(x^2 + y^2 - 3xy) \\ &= (x^2 + 3xy + y^2)(x^2 - 3xy + y^2) \end{aligned}$$

#### 2. Factorize each of the following expressions:

(i)  $(x + 1)(x + 2)(x + 3)(x + 4) + 1$

$$(x + 1)(x + 2)(x + 3)(x + 4) + 1$$

$$1 + 4 = 5 = 2 + 3$$

$$\begin{aligned} &= [(x + 1)(x + 4)][(x + 2)(x + 3)] + 1 \\ &= [x(x + 4) + 1(x + 4)][x(x + 3) + 2(x + 3)] + 1 \\ &= [x^2 + 4x + x + 4][x^2 + 3x + 2x + 6] + 1 \\ &= [x^2 + 5x + 4][x^2 + 5x + 6] + 1 \end{aligned}$$

Let  $y = x^2 + 5x$

$$\begin{aligned} &= (y + 4)(y + 6) + 1 \\ &= y(y + 6) + 4(y + 6) + 1 \\ &= y^2 + 6y + 4y + 24 + 1 \\ &= y^2 + 10y + 25 \\ &= y^2 + 5y + 5y + 25 \\ &= y(y + 5) + 5(y + 5) \\ &= (y + 5)(y + 5) \\ &= (y + 5)^2 \\ &= (x^2 + 5x + 5)^2 \end{aligned}$$

(ii)  $(x + 2)(x - 7)(x - 4)(x - 1) + 17$

$$(x + 2)(x - 7)(x - 4)(x - 1) + 17$$

$$2 - 7 = -5 = -4 - 1$$

$$\begin{aligned} &= [(x + 2)(x - 7)][(x - 4)(x - 1)] + 17 \\ &= [x(x - 7) + 2(x - 7)][x(x - 1) - 4(x - 1)] + 17 \\ &= [x^2 - 7x + 2x - 14][x^2 - x - 4x + 4] + 17 \\ &= [x^2 - 5x - 14][x^2 - 5x + 4] + 1 \end{aligned}$$

Let  $y = x^2 - 5x$

$$\begin{aligned} &= (y - 14)(y + 4) + 17 \\ &= y(y + 4) - 14(y + 4) + 17 \\ &= y^2 + 4y - 14y - 56 + 17 \\ &= y^2 - 10y - 39 \\ &= y^2 - 13y + 3y - 39 \\ &= y(y - 13) + 3(y - 13) \\ &= (y - 13)(y + 3) \\ &= (x^2 - 5x - 13)(x^2 - 5x + 3) \end{aligned}$$

$$\text{(iii)} (2x^2 + 7x + 3)(2x^2 + 7x + 5) + 1$$

$$(2x^2 + 7x + 3)(2x^2 + 7x + 5) + 1$$

$$\text{Let } y = 2x^2 + 7x$$

$$= (y + 3)(y + 5) + 1$$

$$= y(y + 5) + 3(y + 5) + 1$$

$$= y^2 + 5y + 3y + 15 + 1$$

$$= y^2 + 8y + 16$$

$$= y^2 + 4y + 4y + 16$$

$$= y(y + 4) + 4(y + 4)$$

$$= (y + 4)(y + 4)$$

$$= (y + 4)^2$$

$$= (2x^2 + 7x + 4)^2$$

$$\text{(iv)} (3x^2 + 5x + 3)(3x^2 + 7x + 5) - 3$$

$$(3x^2 + 5x + 3)(3x^2 + 7x + 5) - 3$$

$$\text{Let } y = 3x^2 + 5x$$

$$= (y + 3)(y + 5) - 3$$

$$= y(y + 5) + 3(y + 5) - 3$$

$$= y^2 + 5y + 3y + 15 - 3$$

$$= y^2 + 8y + 12$$

$$= y^2 + 6y + 2y + 12$$

$$= y(y + 6) + 2(y + 6)$$

$$= (y + 6)(y + 2)$$

$$= (3x^2 + 5x + 6)(3x^2 + 5x + 2)$$

$$\text{(v)} (x + 1)(x + 2)(x + 3)(x + 6) - 3x^2$$

$$(x + 1)(x + 2)(x + 3)(x + 6) - 3x^2$$

$$1 \times 6 = 6 = 2 \times 3$$

$$= [(x + 1)(x + 6)][(x + 2)(x + 3)] - 3x^2$$

$$= [x(x + 6) + 1(x + 6)][x(x + 3) + 2(x + 3)] - 3x^2$$

$$= [x^2 + 6x + x + 6][x^2 + 3x + 2x + 6] - 3x^2$$

$$= [x^2 + 7x + 6][x^2 + 5x + 6] - 3x^2$$

$$= [x^2 + 6 + 7x][x^2 + 6 + 5x] - 3x^2$$

$$\text{Let } y = x^2 + 6$$

$$= (y + 7x)(y + 5x) - 3x^2$$

$$= y(y + 5x) + 7x(y + 5x) - 3x^2$$

$$= y^2 + 5xy + 7xy + 35x^2 - 3x^2$$

$$= y^2 + 12xy + 32x^2$$

$$= y^2 + 8xy + 4xy + 32x^2$$

$$= y(y + 8x) + 4x(y + 8x)$$

$$= (y + 8x)(y + 4x)$$

$$= (x^2 + 6 + 8x)(x^2 + 6 + 4x)$$

$$= (x^2 + 8x + 6)(x^2 + 4x + 6)$$

$$\text{(vi)} (x + 1)(x - 1)(x + 2)(x - 2) - 16x^2$$

$$(x + 1)(x - 1)(x + 2)(x - 2) - 16x^2$$

$$1 \times 2 = 2 = -1 \times -2$$

$$= [(x + 1)(x + 2)][(x - 1)(x - 2)] - 16x^2$$

$$= [x(x + 2) + 1(x + 2)][x(x - 2) - 1(x - 2)] - 16x^2$$

$$= [x^2 + 2x + x + 2][x^2 - 2x - x + 2] - 16x^2$$

$$= [x^2 + 3x + 2][x^2 - 3x + 2] - 16x^2$$

$$= [x^2 + 2 + 3x][x^2 + 2 - 3x] - 16x^2$$

$$\text{Let } y = x^2 + 2$$

$$= (y + 3x)(y - 3x) - 16x^2$$

$$= y(y - 3x) + 3x(y - 3x) - 16x^2$$

$$= y^2 - 3xy + 3xy - 9x^2 - 16x^2$$

$$= y^2 - 25x^2$$

$$= (y)^2 - (5x)^2$$

$$= (y + 5x)(y - 5x)$$

$$= (x^2 + 2 + 5x)(x^2 + 2 - 5x)$$

$$= (x^2 + 5x + 2)(x^2 - 5x + 2)$$

### 3. Factorize:

$$\text{(i)} 8x^3 + 12x^2 + 6x + 1$$

$$8x^3 + 12x^2 + 6x + 1$$

$$= (2x)^3 + 3(2x)^2(1) + 3(2x)(1)^2 + (1)^3$$

$$= (2x + 1)^3$$

$$\text{(ii)} 27a^3 + 108a^2b + 144ab^2 + 64b^3$$

$$27a^3 + 108a^2b + 144ab^2 + 64b^3$$

$$= (3a)^3 + 3(3a)^2(4b) + 3(3a)(4b)^2 + (4b)^3$$

$$= (3a + 4b)^3$$

$$\text{(iii)} x^3 + 18x^2y + 108xy^2 + 216y^3$$

$$x^3 + 18x^2y + 108xy^2 + 216y^3$$

$$= (x)^3 + 3(x)^2(6y) + 3(x)(6y)^2 + (6y)^3$$

$$= (x + 6y)^3$$

$$\text{(iv)} 8x^3 - 125y^3 + 150xy^2 - 60x^2y$$

$$8x^3 - 125y^3 + 150xy^2 - 60x^2y$$

$$= 8x^3 - 60x^2y + 150xy^2 - 125y^3$$

$$= (2x)^3 - 3(2x)^2(5y) + 3(2x)(5y)^2 + (5y)^3$$

$$= (2x - 5y)^3$$

### 3. Factorize:

$$\text{(i)} 125a^3 - 1$$

$$125a^3 - 1$$

$$= (5a)^3 - (1)^3$$

$$= (5a - 1)[(5a)^2 + (5a)(1) + (1)^2]$$

$$= (5a - 1)(25a^2 + 5a + 1)$$

$$\text{(ii)} 64x^3 + 125$$

$$64x^3 + 125$$

$$= (4x)^3 + (5)^3$$

$$= (4x + 5)[(4x)^2 + (4x)(5) + (5)^2]$$

$$= (4x + 5)(16x^2 + 20x + 25)$$

$$\text{(iii)} x^6 - 27$$

$$x^6 - 27$$

$$= (x^2)^3 - (3)^3$$

$$= (x^2 - 3)[(x^2)^2 + (x^2)(3) + (3)^2]$$

$$= (x^2 - 3)(x^4 + 3x^2 + 9)$$

**(iv)  $1000a^3 + 1$**

$$\begin{aligned} & 1000a^3 + 1 \\ &= (10a)^3 + (1)^3 \\ &= (10a + 1)[(10a)^2 + (10a)(1) + (1)^2] \\ &= (10a + 1)(100a^2 + 10a + 1) \end{aligned}$$

**(v)  $343x^3 + 216$**

$$\begin{aligned} & 343x^3 + 216 \\ &= (7x)^3 + (6)^3 \\ &= (7x + 6)[(7x)^2 + (7x)(6) + (6)^2] \\ &= (7x + 6)(49x^2 + 42x + 36) \end{aligned}$$

**(v)  $27 - 512y^3$**

$$\begin{aligned} & 27 - 512y^3 \\ &= (3)^3 - (8y)^3 \\ &= (3 - 8y)[(3)^2 + (3)(8y) + (8y)^2] \\ &= (3 - 8y)(9 + 24y + 64y^2) \end{aligned}$$

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